19 Reliability

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Topic Highlights

Measurements of Successful Operation—No Repair
Useful Approximations
Measurements of Successful Operation—Repairable
Average Unavailability with Periodic Inspection and Test
Periodic Restoration and Imperfect Testing
Equipment Failure Modes
SIF Modeling of Failure Modes
Redundancy

19.1 Introduction

There are a number of common metrics used within the field of reliability engineering. Primary ones include reliability, unreliability, availability, unavailability and mean time to failure (MTTF). But, when different failure modes are considered—as they are when doing safety integrated functions (SIF) verification—then new metrics are needed. These include probability of failing safely (PFS), probability of failure on demand (PFD), probability of failure on demand average (PFDavg), mean-time-to-failure spurious ($MTTF_S$), and mean time to dangerous failure ($MTTF_D$).

19.2 Measurements of Successful Operation – No Repair

Probability of Success—This is often defined as the probability that a system will perform its intended function, when needed, and when operated within its specified limits. The phrase at the end of the last sentence tells the user of the equipment that the published failure rates only apply when the system is not abused or otherwise operated outside its specified limits.

Using the rules of reliability engineering, one can calculate probability of successful operation for a particular set of circumstances. Depending on the circumstances, that probability is called "reliability" or "availability" (or, on occasion, some other name).

Reliability—a measure of successful operation for a specified interval of time. Reliability, R(t), is defined as "the probability that a system will perform its intended function when required to do so if operated within its specified limits for a specified operating time interval." The definition includes five important aspects:

1. The system's "intended function" must be known.
2. "When the system is required to function" must be judged.

3. "Satisfactory performance" must be determined.
4. The "specified design limits" must be known.
5. An operating time interval is specified.

Consider a newly manufactured and successfully tested component. It operates properly when put into service (T = 0). As the operating time interval (T) increases, it becomes less likely that the component will remain successful. Since the component will eventually fail, the probability of success for an infinite time interval is zero. Thus, all reliability functions start at a probability of one and decrease to a probability of zero (Figure 19-1).

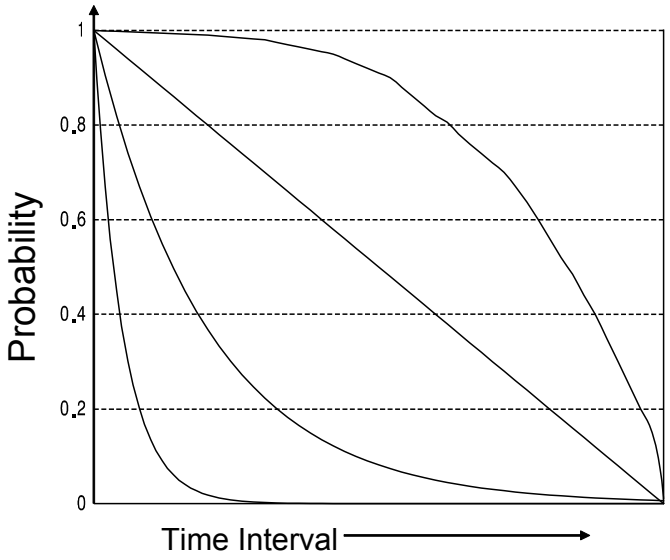


Figure 19-1: Example Reliability Plots

Reliability is a function of the operating time interval. A statement such as "system reliability is 0.95" is meaningless because the time interval is not known. The statement "The reliability equals 0.98 for a mission time of one hundred hours" makes perfect sense.

A reliability function can be derived directly from probability theory. Assume the probability of successful operation for a one-hour time interval is 0.999. What is the probability of successful operation for a two-hour time interval? The system will be successful only if it is successful for both the first hour and the second hour. Therefore the two hour probability of success equals:

$$0.999 \times 0.999 = 0.998 \tag{19-1}$$

The analysis can be continued for longer time intervals. For each time interval the probability can be calculated by the equation:

$$P(t) = 0.999^t \tag{19-2}$$

Figure 19-2 shows a plot of probability versus operating time using this equation. The plot is a reliability function.

Reliability is a metric originally developed to determine probability of successful operation for a specific "mission time." For example: if a flight time is 10 hours, a logical question is, "What is the probability of successful operation for the entire flight?" The answer would be the *Reliability* for the 10 hours

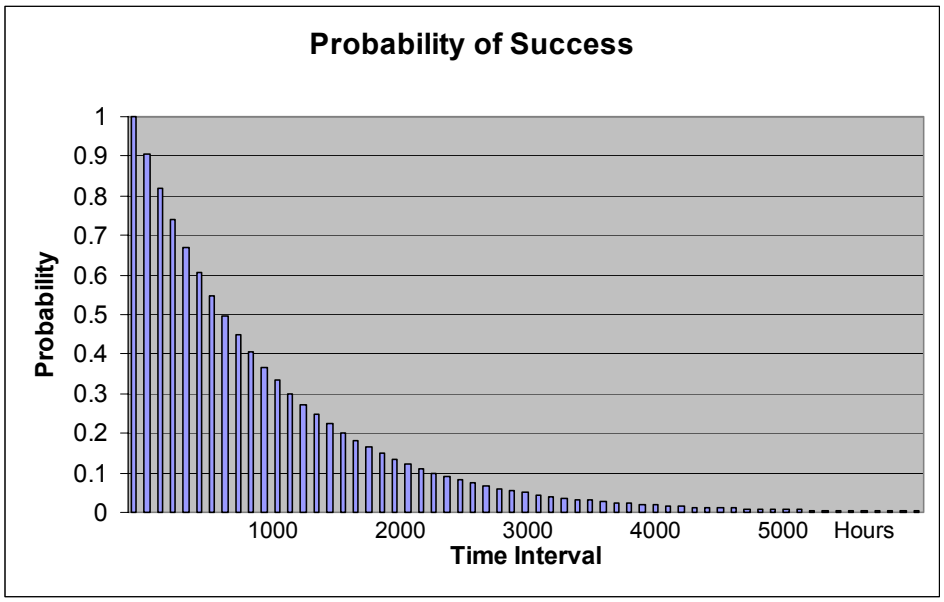


Figure 19-2: Reliability Plot Using Constant Probability of Failure

duration. It is generally a measurement applicable to situations where online repair is not possible, like an unmanned space flight or an airborne aircraft. Unreliability is the complement of reliability. It is defined as the probability of failure during a specific mission time.

Mean Time To Failure (MTTF)—One of the most widely used reliability parameters is the MTTF. It has been formally defined as the "expected value" of the random variable *Time To Fail, T*. Unfortunately, the metric has evolved into a confusing number. MTTF has been misused and misunderstood. It has been misinterpreted as "guaranteed minimum life."

Formulas for MTTF are derived and often used for products during the useful life period. This method excludes wearout failures. Ask an experienced plant engineer "What is the MTTF of a pressure transmitter?" This engineer would likely include wearout and might answer "35 years." Then the engineer would look at the specified MTTF of 300 years and think that the person who calculated that number should come out and stay with him for a few years and see the real world.

Generally, the term MTTF is being defined during the useful life of a device. "End of life" failures are generally not included in the number.

Constant Failure Rate—When a constant failure rate is assumed (which is valid during the useful life of a device) then the relationship between reliability, unreliability, and MTTF are straightforward. If the failure rate is constant then:

$$\lambda(t) = \lambda \tag{19-3}$$

For that assumption it can be shown that:

$$R(t) = e^{-\lambda t} \tag{19-4}$$

$$F(t) = 1 - e^{-\lambda t} \tag{19-5}$$

And

$$MTTF = 1/\lambda \tag{19-6}$$

Figure 19-3 shows the reliability and unreliability functions for a constant failure rate of 0.001 failures per hour. Note the plot for reliability looks the same as Figure 19-2, which shows the probability of successful operation given a probability of success for one hour of 0.999. It can be shown that a constant probability of success is equivalent to an exponential probability of success distribution as a function of operating time interval.

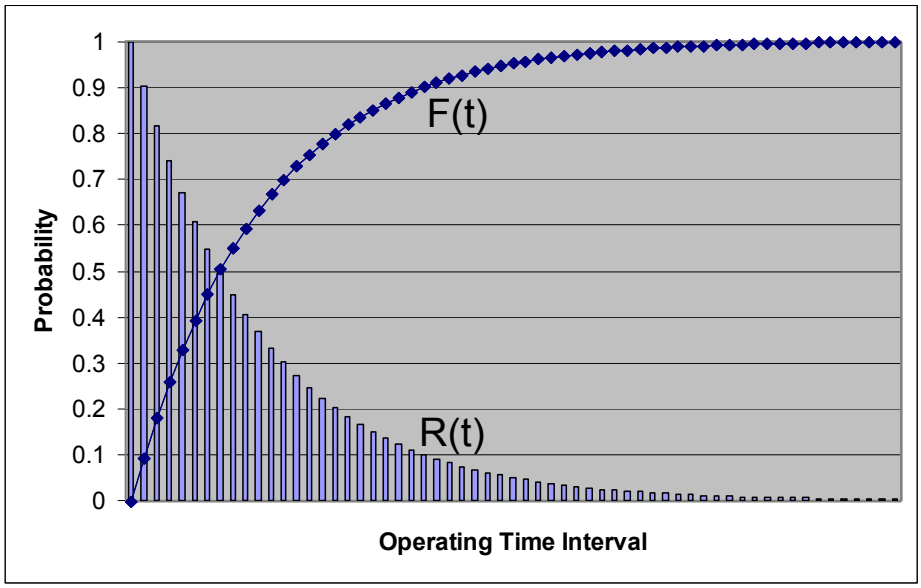


Figure 19-3: Reliability/Unreliability Function for a Constant Failure Rate

19.3 Useful Approximations

Mathematically, it can be shown certain functions can be approximated by a series of other functions. For all values of *x*, it can be shown that

$$e^x = 1 + x + x^2/2! + x^3/3! + x^4/4! + \ldots \tag{19-7}$$

For a sufficiently small value of *x*, the exponential can be approximated with

$$e^x = 1 + x$$

Substituting $-\lambda t$ for x

$$e^{\lambda t} = 1 + \lambda t$$

Thus, there is an approximation for unreliability when λt is sufficiently small

$$F(t) = \lambda t \tag{19-8}$$

Remember, this is only an approximation and not a fundamental equation. Often the notation for unreliability is PF (probability of failure) and the equation is shown as

$$PF(t) = \lambda t \tag{19-9}$$

19.4 Measurements of Successful Operation—Repairable Systems

The measurement "reliability" requires that a system be successful for an interval of time. While this probability is a valuable estimate for situations where a system cannot be repaired during a mission, something different is needed for an industrial process control system where repairs can be made—often with the process operating.

Mean Time To Restore (MTTR)—MTTR is the "expected value" of the random variable "restore time" (or time to repair). The definition includes the time required to detect that a failure has occurred as well as the time required to make a repair once the failure has been detected and identified. Like MTTF, MTTR is an average value. MTTR is the average time required to move from unsuccessful operation to successful operation.

In the past, the acronym MTTR stood for "Mean Time To Repair." The term was changed in IEC 61508 because of confusion as to what was included. Some thought that "Mean Time To Repair" included only actual repair time. Others interpreted the term to include both time to detect a failure (diagnostic time) and actual repair time. The term "Mean Dead Time (MDT)" is commonly used in some parts of the world and means the same as Mean Time To Restore.

Mean Time To Restore (MTTR) is a term created to clearly include both diagnostic detection time and actual repair time. Of course, when actually estimating MTTR, one must include time to detect, recognize and identify the failure, time to obtain spare parts, time for repair team personnel to respond, actual time to do the repair, time to document all activities and time to get the equipment back in operation.

Reliability engineers often make the assumption that the probability of repair is an exponentially distributed function, in which case the "restore rate" is a constant. The lower case Greek letter mu is used to represent restore rate by convention. The equation for restore rate is:

$$\mu = 1/\text{MTTR} \tag{19-10}$$

Restore times can be difficult to estimate. This is especially true when periodic activities are involved. Imagine the situation where a failure in the safety instrumented system is not noticed until a periodic inspection and test is done. The failure may occur right before the inspection and test, in which case the detection time might be near zero. Or it may occur right after the inspection and test, in which case the detection time may get as large as the inspection period.

In such cases, it is probably best to model repair probability as a periodic function, not as a constant (see Ref. 1). This is discussed later in the section "Average Unavailability with Periodic Inspection and Test."

Mean Time Between Failures (MTBF)—MTBF is defined as the average time period of a failure/repair cycle. It includes time to failure, any time required to detect the failure and actual repair time. This implies that a component has failed and then has been successfully repaired. For a simple repairable component,

$$\text{MTBF} = \text{MTTF} + \text{MTTR} \tag{19-11}$$

The MTBF term can also be confusing. Since MTTR is usually much smaller than MTTF, MTBF is approximately equal to MTTF. The term MTBF is often substituted for MTTF and applies to both repairable systems and non-repairable systems.

Availability—The reliability measurement was not sufficiently useful for engineers who needed to know the average chance of success of a system when repairs are possible. Another measure of system success for repairable systems was needed. That metric is "availability." Availability is defined as "the probability that a device is successful at time t when needed and operated within specified limits." No

operating time interval is directly involved. If a system is operating successfully, it is available. It does not matter whether it has failed in the past and has been repaired or has been operating continuously from startup without any failures. Availability is a measure of "uptime" in a system, unit, or module.

Availability and reliability are different metrics. Reliability is always a function of failure rates and operating time interval. Availability is a function of failure rates and repair rates. While instantaneous availability will vary during the operating time interval, this is due to changes in failure probabilities and repair situations. Often availability is calculated as an average over a long operating time interval. This is referred to as "steady state availability."

In some systems, especially safety instrumented systems (SIS), the repair situation is not constant. In safety instrumented systems the situation occurs when failures are discovered and repaired during a periodic inspection and test. For these systems, steady state availability is NOT a good measure of system success. Instead, average availability is calculated for the operating time interval between inspections. [Note: This is not the same measurement as steady state availability.]

Unavailability—A measure of failure that is used primarily for repairable systems. It is defined as "the probability that a device is not successful (is failed) at time t." Different metrics can be calculated including steady state unavailability and average unavailability over an operating time interval. Unavailability is the one's complement of availability; therefore,

$$U(t) = 1 - A(t) \tag{19-12}$$

Steady State Availability—Traditionally, reliability engineers have assumed a constant repair rate. When this is done, probability models can be solved for "steady state" or average probability of successful operation. The metric can be useful, but it has relevance only for long intervals of time.

Figure 19-4 shows a Markov probability model of a single component with a single failure mode. This model can be solved for steady state availability and steady state unavailability.

$$A = MTTF/(MTTF + MTTR) \tag{19-13}$$

$$U = MTTR/(MTTF + MTTR) \tag{19-14}$$

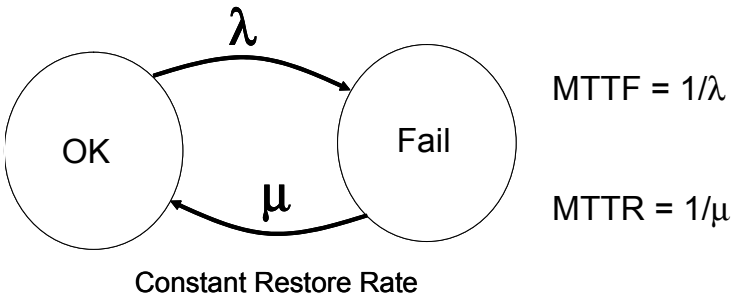


Figure 19-4: Markov Model for Single Component and Single Failure Mode

When the Markov model of Figure 19-4 is solved for availability as a function of operating time interval, the result is shown in Figure 19-5 labeled A(t). It can be seen that the availability reaches a "steady state" after some period of time.

Figure 19-6 shows a plot of Unavailability versus Unreliability. These plots are complementary to those shown in Figure 19-5.

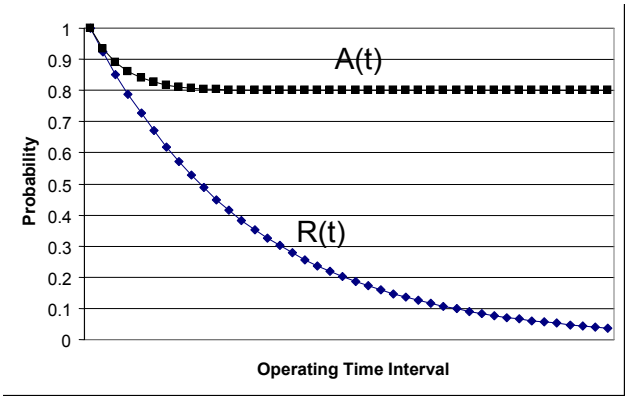


Figure 19-5: Reliability Versus Availability

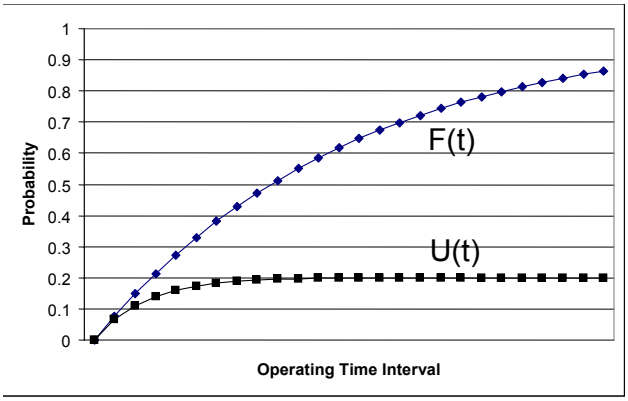


Figure 19-6: Unreliability Versus Unavailability

19.5 Average Unavailability with Periodic Inspection and Test

In low demand safety instrumented system applications, the restore rate is NOT constant. For failures not detected until a periodic inspection and test, the restore rate is zero until the time of the test. If it is discovered the system is operating successfully, then the probability of failure is set to zero. If it is discovered the system has a failure, it is repaired. In both cases the restore rate is high for a brief period of time. Dr. Julia V. Bukowski has described this situation and proposed modeling repair as a periodic impulse function (Ref. 1).

Figure 19-7 shows a plot of probability of failure in this situation. This can be compared with unavailability calculated with the constant restore rate model as a function of operating time. With the constant restore model, the unavailability reaches a steady state value. This value is clearly different than the result that would be obtained by averaging the unavailability calculated using a periodic restore period.

It is often assumed that periodic inspection and test will detect all failed components, and the system will be renewed to perfect condition. Therefore, the unreliability function is suitable for the problem. A mission time equal to the time between periodic inspection and test is used. In safety instrumented system applications, the objective is to find a model for the probability that a system will fail when a dangerous condition occurs. This dangerous condition is called a “demand.”

Our objective, then, is to calculate the probability of failure on demand. If the system is operating in an environment where demands are infrequent (for example, once per 10 years) and independent from

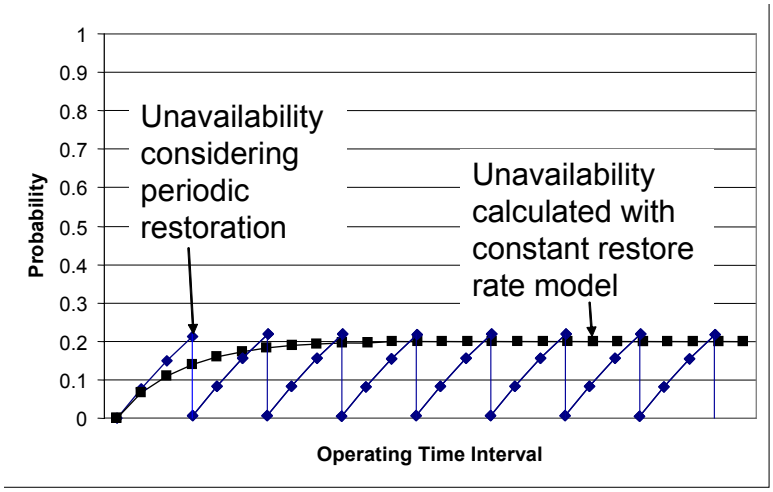


Figure 19-7: Probability of Failure with Periodic Testing

system proof tests; then an average of the unreliability function will provide the average probability of failure. This by definition is an "unavailability function" since repair is allowed. [Note: This averaging technique is not valid when demands are more frequent. Special modeling techniques are needed in that case.]

As an example, consider the single component unreliability function given in Equation 19-5.

$$F(t) = 1 - e^{-\lambda t}$$

This can be approximated as explained previously with Equation 19-8.

$$F(t) = \lambda t$$

The average can be obtained by using the expected value equation

$$PFavg = \frac{1}{T}\int_0^T PF(t)dt \tag{19-15}$$

with the result being an approximation equation

$$PFavg = \lambda t/2 \tag{19-16}$$

For a single component (non-redundant) or a single channel system; the approximation is shown in Figure 19-8. Note that the approximation is conservative and supplies a pessimistic value.

19.6 Periodic Restoration and Imperfect Testing

It is quite unrealistic to assume that inspection and testing processes will detect all failures. In the worst case, assume that testing is not done. In that situation what is the mission time? If the equipment is used for the life of an industrial facility then plant life is the mission time. Probability of failure would be modeled with the unreliability function using the plant life as the time interval.

If the equipment is required to operate only on demand, and the demand is independent of system failure, then the unreliability function can be averaged as explained in the preceding section.

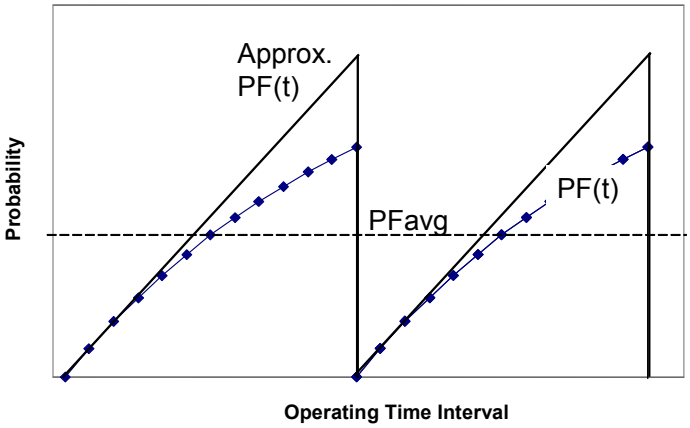


Figure 19-8: Average Probability of Failure Approximation with Periodic Testing

When only some failures are detected during the periodic inspection and test, then the average probability of failure can be calculated using an equation that combines the two types of failures—those detected by the test and those undetected by the test. One must estimate the percentage of failures detected by the test to make this split. The equation would be:

$$PFavg = C_{PT} \lambda TI/2 + (1 - C_{PT}) \lambda LT/2 \quad (19\text{-}17)$$

where

λ = the failure rate
C_{PT} = the percentage of failures detected by the proof test
TI = the periodic test interval
LT = lifetime of the process unit

19.7 Equipment Failure Modes

Instrumentation equipment can fail in different ways. We call these “failure modes.” Consider a two wire pressure transmitter. This instrument is designed to provide a 4–20 milliamp electrical current signal in proportion to the pressure input. Detail failure modes, effects and diagnostic analysis of several of these devices reveals a number of failure modes; frozen output, current to upper limit, current to lower limit, diagnostic failure, communications failure and drifting/erratic output among perhaps others. These instrument failures can be classified into failure mode categories when the application is known.

If a single transmitter (no redundancy) were connected to a safety PLC programmed to trip when the current goes up (high trip), then the instrument failure modes could be classified as shown in Table 19-1.

Table 19-1: Transmitter Failure Mode Categories

Instrument Failure Mode	SIF Failure Mode
Frozen output	Fail-Danger
Output to upper limit	Fail-Safe
Output to lower limit	Fail-Danger
Diagnostic failure	Annunciation
Communication failure	No Effect
Drifting / erratic output	Fail-Danger

Consider possible failure modes of a PLC with a digital input and a digital output, both in a de-energize to trip (logic 0) design. The PLC failure modes can be categorized relative to the safety function as shown in Table 19-2.

Table 19-2: PLC Failure Mode Categories

Instrument Failure Mode	SIF Failure mode
Input stuck High	Fail-Danger
Input stuck low	Fail-Safe
Input circuit oscillates	Fail-Danger*
Output stuck high	Fail-Danger
Output stuck low	Fail-Safe
Improper CPU execution	50% Fail-Safe
	50% Fail-Danger
Memory transient failure	50% Fail-Safe
	50% Fail-Danger
Memory permanent failure	50% Fail-Safe
	50% Fail-Danger
Power supply low (out of tolerance)	Fail-Danger*
Power supply high (out of tolerance)	Fail-Danger*
Power supply zero	Fail-Safe
Diagnostic timer failure	Annunciation
Loss of communication link	No Effect
Display panel failed	No Effect
* unpredictable - assume worst case	

Final element components will fail also and, again, the specific failure modes of the components can be classified into relevant failure modes depending on the application. It is important to know if a valve will open or close on trip. Table 19-3 shows an example failure mode classification based on a close to trip configuration.

Table 19-3: Final Element Failure Mode Categories

Instrument Failure Mode	SIF Failure mode
Solenoid plunger stuck	Fail-Danger
Solenoid coil burnout	Fail-Safe
Actuator shaft failure	Fail-Danger*
Actuator seal failure	Fail-Safe
Actuator spring failure	Fail-Danger
Actuator structure failure - air	Fail-Safe
Actuator structure failure - binding	Fail-Danger*
Valve shaft failure	Fail-Danger*
Valve external seal failure	No Effect
Valve internal seal damage	Fail-Danger
Valve ball stuck in position	Fail-Danger
* unpredictable - assume worst case	

It should be noted the above failure mode categories apply to an individual instrument and may not apply to the set of equipment that performs a safety instrumented function, as the equipment set may contain redundancy. It should be also made clear that the above listings are not intended to be comprehensive or representative of all component types.

Fail-Safe

Most practitioners use a definition of "Fail-Safe" for an instrument to define *a failure that causes a "false or spurious" trip of a safety instrumented function unless that trip is prevented by the architecture of the safety instrumented function.* Many formal definitions have been attempted that include "a failure which causes the system to go to a safe state or increases the probability of going to a safe state." This definition is useful at the system level and includes many cases where redundant architectures are used.

IEC 61508 uses the definition "failure which does not have the potential to put the safety-related system in a hazardous or fail-to-function state." This definition includes many failures that do not cause a false trip under any circumstances and is quite different from the definition practitioners need to calculate the false trip probability.

Fail-Danger

Many practitioners define "Fail-Danger" as *a failure that prevents a safety instrumented function from performing its automatic protection function*. Variations of this definition exist in standards. IEC 61508 provides a definition similar to the one used in this book which reads "failure which has the potential to put the safety-related system in a hazardous or fail-to-function state." The definition from IEC 61508 goes on to add a note: "Whether or not the potential is realized may depend on the channel architecture of the system; in systems with multiple channels to improve safety, a dangerous hardware failure is less likely to lead to the overall dangerous or fail-to-function state." The note from IEC 61508 recognizes that a definition for a piece of equipment may not have the same meaning at the safety instrumented function level or the system level.

Annunciation

Some practitioners recognize that certain failures within equipment used in a safety instrumented function prevent the automatic diagnostics from correct operation. When reliability models are built, many account for the automatic diagnostics ability to reduce the probability of failure. When these diagnostics stop working, the probability of dangerous failure or false trip is increased. While these effects may not be significant, unless they are modeled the effect is not known.

An annunciation failure is therefore defined as *a failure that prevents automatic diagnostics from detecting or annunciating that a failure has occurred inside the equipment***.** Note the failure may be within the equipment that fails or inside an external piece of equipment designed for the purpose of automatic diagnostics. These failures would be classified as "Fail-Safe" in the definition provided in IEC 61508.

No Effect

Some failures within a piece of equipment have no effect on the safety instrumented function nor cause a false trip nor prevent automatic diagnostics from working. Some functionality performed by the equipment is impaired but that functionality is not needed. These may simply be called "No Effect" failures. They are typically not used in any reliability model intended to obtain probability of a false trip or probability of a fail-danger. Per IEC61508, these would be classified as "Fail-Safe" or may be excluded completely from any analysis depending on interpretation of the analyst.

Detected/Undetected

Failure modes can be further be classified as "detected" or "undetected" by automatic diagnostics performed somewhere in the safety instrumented system.

19.8 Safety Integrated Function (SIF) Modeling of Failure Modes

When evaluating safety instrumented function safety integrity, an engineer must examine more than the probability of successful operation. The failure modes of the system must be individually calculated. The normal metrics of reliability, availability and MTTF only suggest a measure of success. Additional metrics to measure safety integrity include probability of failure on demand (PFD), average

probability of failure on demand (PFDavg), risk reduction factor (RRF) and mean time to fail dangerously ($MTTF_D$). Other related terms are probability of failing safely (PFS) and mean time to fail spurious ($MTTF_S$).

PFS/PFD

There is a probability that a safety instrumented function will fail and cause a spurious/false trip of the process. This is called probability of failure safely (PFS). There is also a probability that a safety instrumented function will fail such that it cannot respond to potentially dangerous condition. This is called probability of failure on demand (PFD).

PFDavg

PFD average (PFDavg) is a term used to describe the average probability of failure on demand. PFD will vary as a function of the operating time interval of the equipment. It will not reach a steady state value if any periodic inspection, test and repair is done. Therefore, the average value of PFD over a period of time can be a useful metric if it assumed that the potentially dangerous condition (also called hazard) is independent from equipment failures in the safety instrumented function.

The assumption of independence between hazards and safety instrumented function failures seems very realistic. (NOTE: If control functions and safety functions are performed by the same equipment, the assumption may not be valid! Detailed analysis must be done to ensure safety in such situations, and it is best to avoid such designs completely.) When hazards and equipments are independent, it is realized a hazard may come at any time. Therefore, international standards have specified that PFDavg is an appropriate metric for measuring the effectiveness of a safety instrumented function.

PFDavg is defined as the arithmetic mean over a defined time interval. For situations where a safety instrumented function is periodically inspected and tested, the test interval is correct time period. Therefore:

$$PFDavg(TI) = \frac{1}{TI}\int_0^{TI}(PFD)dt$$

This definition is used to obtain numerical results in several of the system modeling techniques. In a discrete time Markov model using numerical solution techniques, a direct average of the time dependent numerical values will provide the most accurate answer. When analytical equations for PFD are obtained using a fault tree, the above equation can be used to obtain equations for PFDavg.

19.9 Redundancy

There are applications where the reliability or safety integrity of a single instrument is not sufficient. In these cases more than one instrument is used in a design. Some arrangements of the instruments are designed to provide higher reliability (typically to protect against a single "safe" failure). Other arrangements of instruments are designed to provide higher safety integrity (typically to protect against a single "dangerous" failure). And some arrangements are designed to provide both high reliability and high safety integrity. When multiple instruments are wired (or configured) to provide redundancy to protect against one or more failure modes, these arrangements are known as "architectures." A listing of some common architectures is shown in Table 19-4. These architectures are described in detail in Chapter 14 of Reference 2.

Table 19-4: Common Redundant Architectures

Architecture	Number of units	Output Switches	Objective
1oo1	1	1	Base unit
1oo2	2	2	High Safety
2oo2	2	2	Maintain output
1oo1D	1	2	High Safety
2oo3	3	6	Safety and Availability
2oo2D	2	4	Safety and Availability
1oo2D	2	4	Safety and Availability

The naming convention stands for X out of Y where Y is the number of equipment sets in the design and X is the number of equipment sets needed to perform the function. In some advanced architecture names, the term D is added to designate a switch that is controlled by diagnostics to reconfigure the equipment if a failure is detected in one equipment set.

19.10 References

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